Architecture Design Methodology on Huawei Cloud

Alan Turing: The Inventor of Computer Science

Alan Turing
(1912.06–1954.06)

- **Key player in World War II**

  Alan Turing played a major role in breaking the cipher systems used by the German Enigma machine. Churchill once said in his memoirs that "Alan Turing made the single biggest contribution to Allied victory in the war against Nazi Germany."

- **The father of modern computers**

  At the age of 24, Turing came up a world-changing idea: the Turing machine, which first appeared in his paper *On Computable Numbers, with an Application to the Entscheidungsproblem*. The Turing machine explains all the basic principles of computers.

- **The founder of AI**

  The Turing Test is a thought experiment proposed by Turing in 1950, to test whether a machine could show intelligent behavior like a human being. During the test, a tester enters regular text using a keyboard and the text is displayed on the screen. No special commands are involved.

The Bank of England issued the new polymer £50 note featuring Turing on June 23, 2021, to commemorate Turing's contribution to computing and artificial intelligence.

ON COMPUTABLE NUMBERS, WITH AN APPLICATION TO THE ENTSCHEIDUNGSPROBLEM

By A. M. TURING.

[Received 28 May, 1936.—Read 12 November, 1936.]

Turing laid the theory foundation of modern computation in 1937

Can Machines Think?
The concept of machine thinking (Turing Test) was first proposed in 1950.

HISILICON

Huawei Turing Business Unit, HS is making breakthroughs in computing chips and solutions.

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John von Neumann: The Designer of Computer Architecture

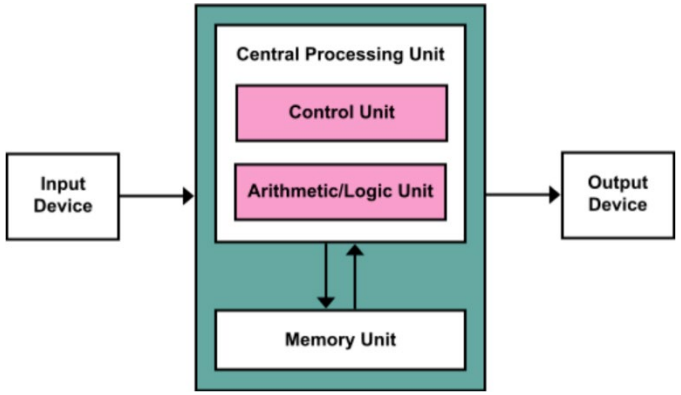
John von Neumann
(1903.12–1957.2)

Since 1945 when John von Neumann proposed this computer architecture, the Von Neumann architecture has always been the way how most general-purpose computers run.

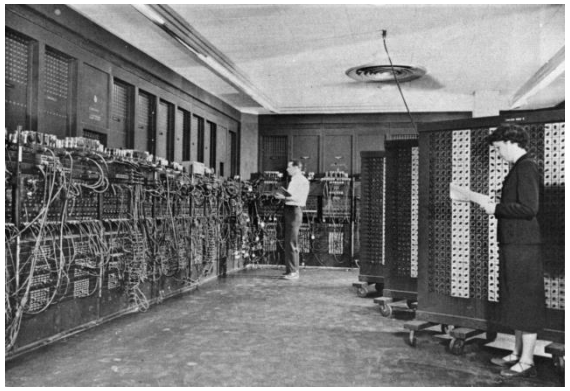
- **Founder of computer architecture**

  John von Neumann was deeply involved in developing the first **programmable electronic computer, the ENIAC**. The ENIAC was able to identify and determine rule sets of other data operations. John von Neumann's most lasting contributions to computer science are the two concepts used in today's every computer: **The Von Neumann architecture and the stored-program concept.** The basic idea of the Von Neumann architecture is as follows:

  - Data and instructions are both stored as binary code. An instruction consists of an operation and an address.
  - Programs and data are stored in a memory unit. The computer retrieves instructions from the memory, executes the instructions, and automatically completes computing tasks. This is the concept of storage-programs and program control.
  - Instructions are executed in the order in which they are stored in memory. Program branches are implemented using transfer instructions.



Von Neumann architecture



Space: 170 m²
Weight: 30 tons
Power consumption: about 150 kW
Computing power: 5,000 operations per second
Purpose: ballistics calculation, U.S. Department of Defense

ENIAC – World's First Electronic Computer

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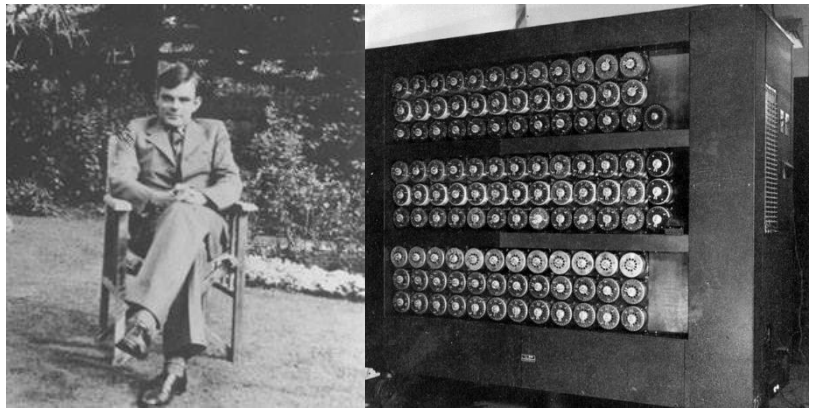
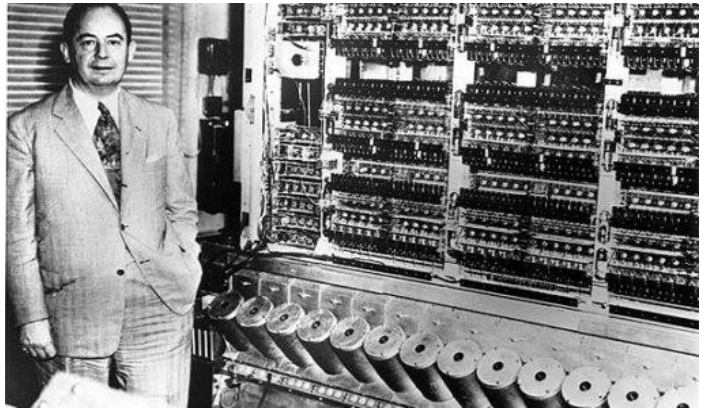
To Commemorate the Great Contributions of the Two Scientists, Related Awards Have Become Top Honors in the Field



In World War II, Turing and his code-breaking machine, Christopher, made outstanding contributions to cracking the Enigma code used by Germany. It is said today that Turing had brought World War II to end at least two years earlier.



In June 1942, John von Neumann started working on the Manhattan Project. His ENIAC computer was used for bombing simulation computing, helping win the key battles of World War II in 1945.

Turing Award

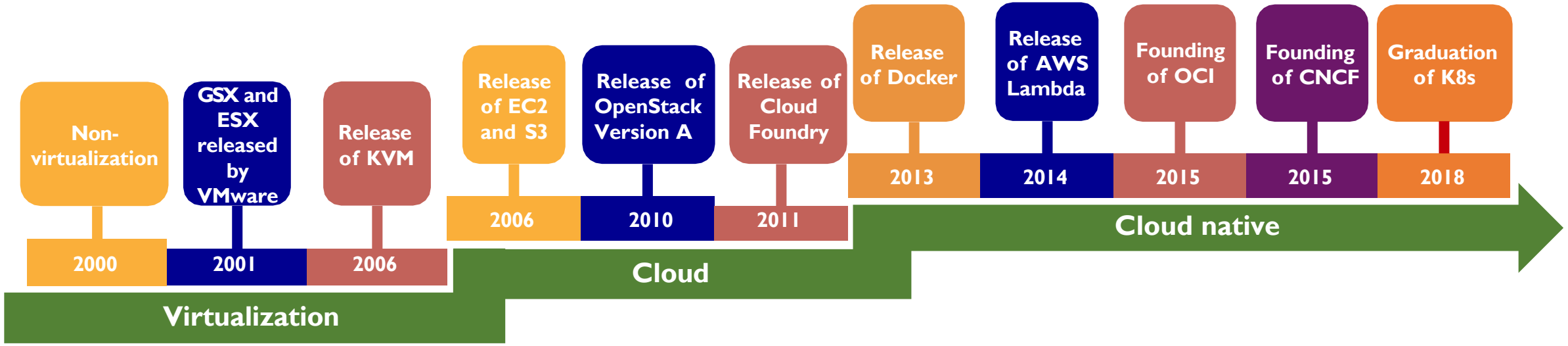
The Turing Award was established by the Association for Computing Machinery (ACM) in 1966 to reward individuals who have made significant contributions to computer science.

John von Neumann Award

The John von Neumann Award was established by the Institute of Electrical and Electronics Engineers (IEEE) in 1990 to reward scientists who have outstanding achievements in computer science and technology.

Cloud Basics and Development Timeline

- Cloud computing refers to providing compute resources through on-demand services over the Internet. Enterprises do not need to purchase, configure, or manage resources on their own, and only need to pay for the resources they use.



Why Cloud Computing?

Cloud computing is the new infrastructure for a digital era. It is a foundation for diverse industries

Mobile payments

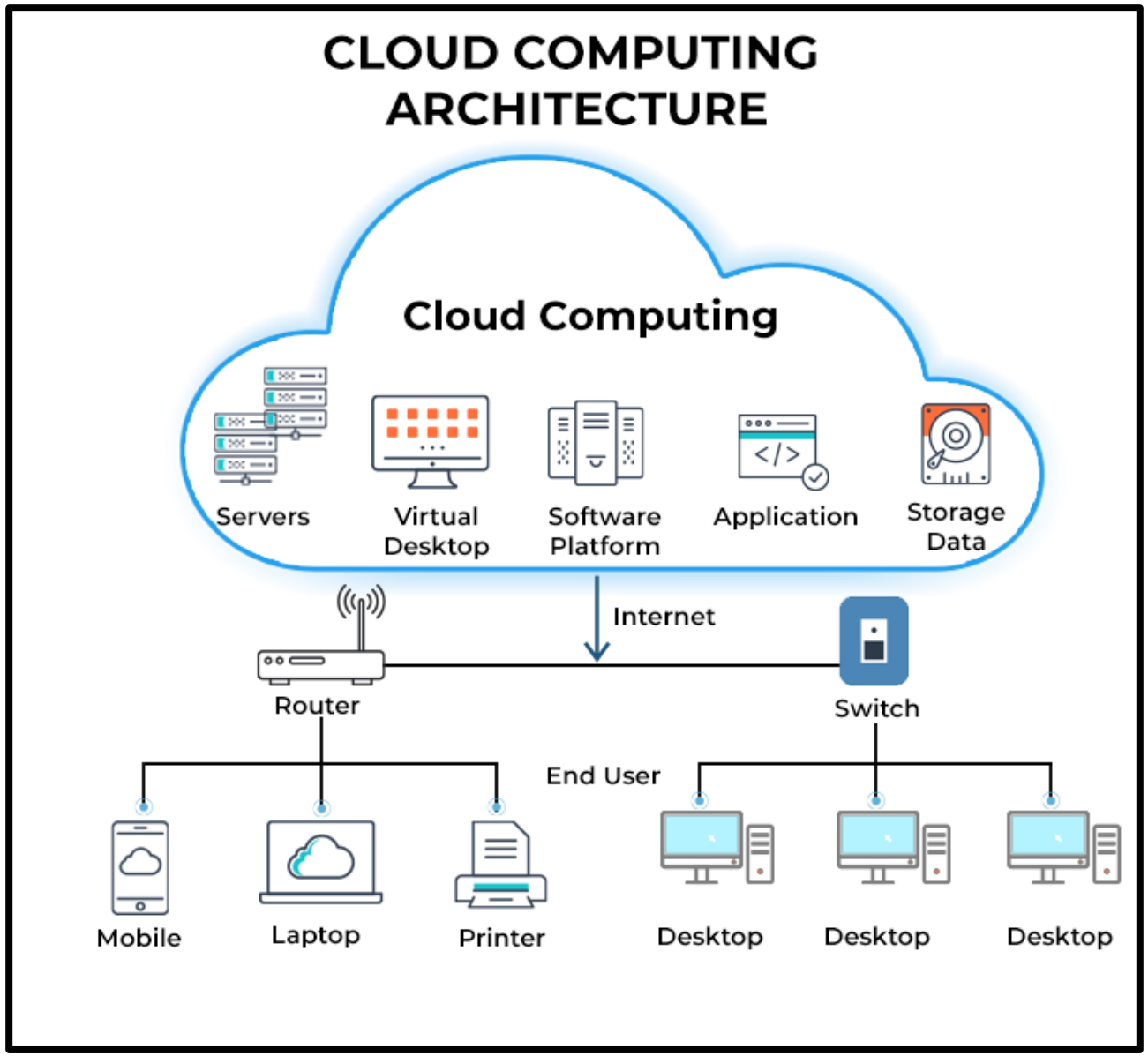
Citizen services

Online car-hailing platforms

Map services

Social platforms

Huawei Mobile Services



What Is Cloud Computing? Definition, Benefits, Types, and Trends - Spiceworks

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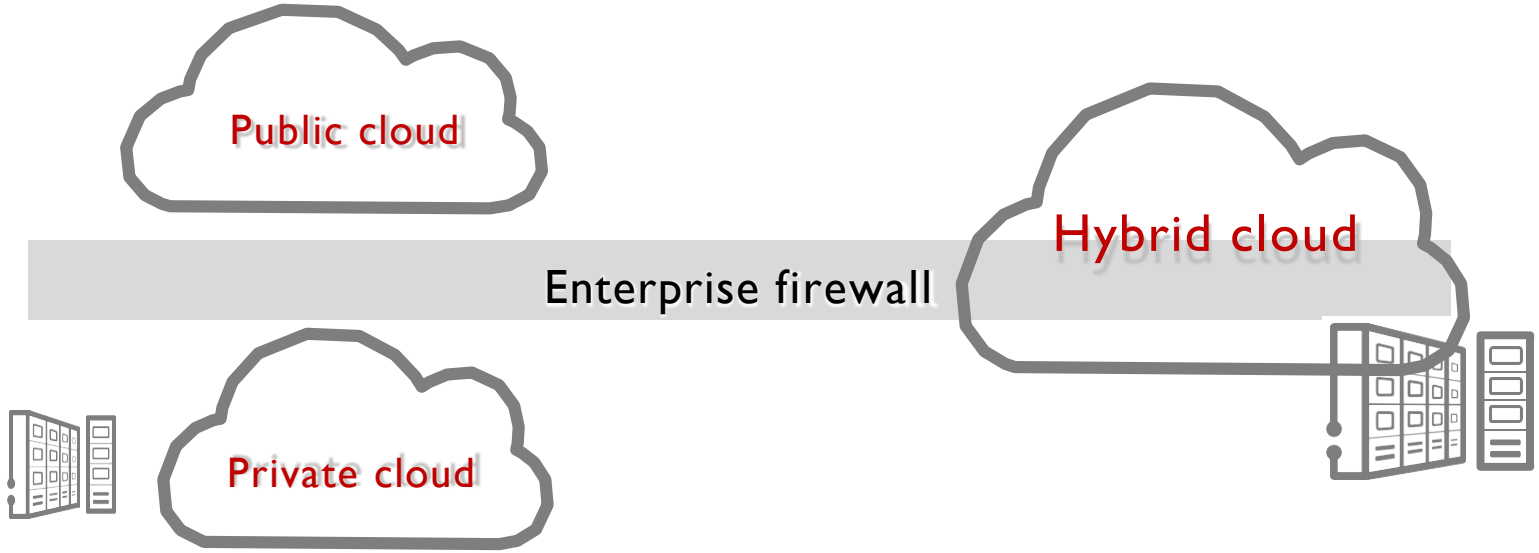
Five Key Characteristics of Cloud Computing

NIST
National Institute of Standards and Technology
U.S. Department of Commerce

- On-demand, self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measurable usage

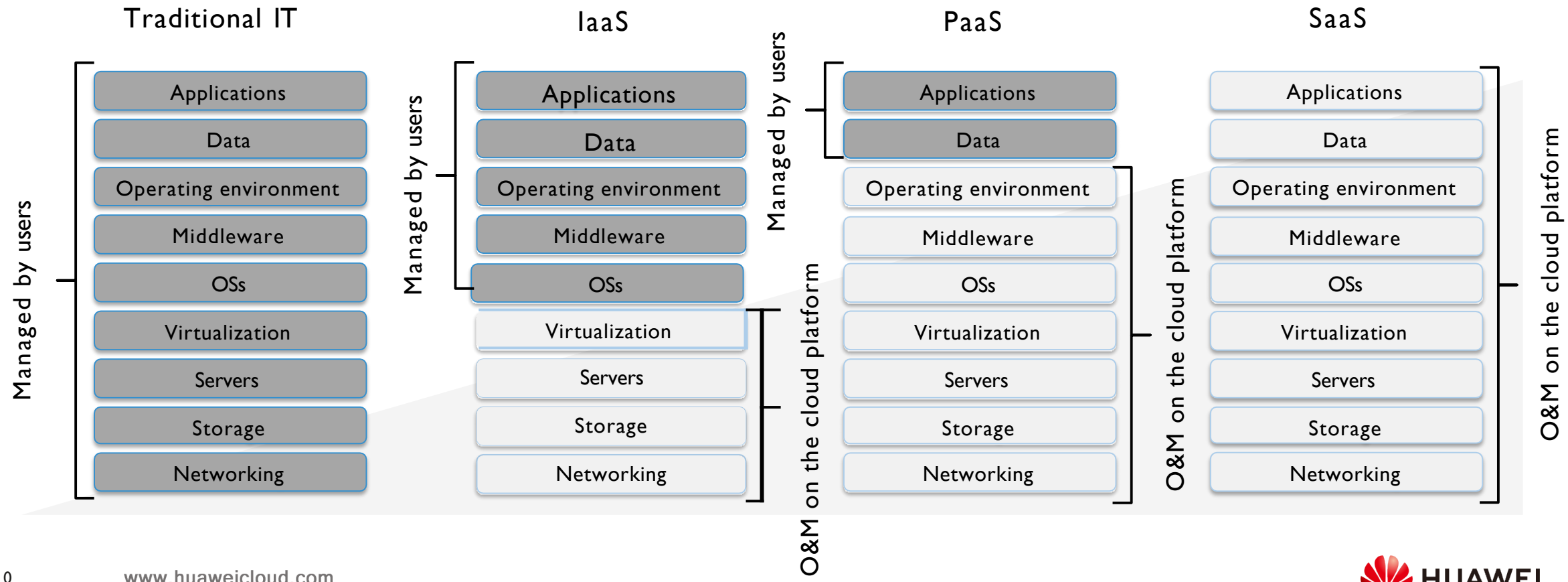
Cloud Computing Development Models



Private cloud: The cloud infrastructure is owned and managed for exclusive use by a single organization.
Public cloud: The cloud service provider owns and operates the cloud infrastructure and provides cloud services open to the public or enterprise customers.
Hybrid cloud: A combination of public and private clouds, viewed as a single cloud externally.

Cloud Computing Service Models

- There are three service models for cloud computing: Infrastructure as a Service (IaaS) that provides compute and storage services, Platform as a Service (PaaS) that provides environments for developing and deploying cloud applications, and Software as a Service (SaaS) that provides applications as services.



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Cloud Computing: A New Computing Model and a New Economic Model

- Cloud computing is the **on-demand delivery** of cloud resources, such as compute, storage, and applications, over the Internet **anytime, anywhere**.

Traditional IT construction = **Digging your own well**

If you want to drink water, you first have to buy a lot of tools and spend a lot of manpower to dig a well, and then you can drink the water.

Cloud computing = **Water on tap**

A cloud service provider is like a water utility that has already built out their own piping system. You just pay for the water you use every month. There is no need to dig your own well or add plumbing.

Cloud service provider

- Cloud service models based on service layers: Cloud computing can be classified into infrastructure as a service (**IaaS**), platform as a service (**PaaS**), and software as a service (**SaaS**).

IaaS is like the **pipes** (servers, storage, and networks) that bring the water to your home.

PaaS is like a **kettle** (development environments and databases) provided for users.

aPaaS: mocha pot
tPaaS: capsule coffee machine

SaaS is like a bottle of **drink**. Users can drink it immediately after opening the bottle.

- Cloud types from the user's perspective: **public cloud, private cloud, and hybrid cloud**

① The user digs their own well: This is like a private cloud, and the well is an **on-premises data center.**

② The user uses tap water: This is like a **public cloud**, and the water utility is a **cloud data center**.

③ The user uses tap water for cooking but well water for bathing: This is like a **hybrid cloud.**

Cloud Computing Benefits

Reduced investment into fixed assets

- Investments into fixed assets are reduced in the early stage so that enterprise resources can be used to tap into the core value of businesses.
- Fixed assets may become a burden for future transformation. For example, business adjustments may be limited by the reuse of existing assets.

Large-scale resources

- Economies of scale, that is, the procurement volume from Huawei is enormous and the price is lower.
- Huawei Cloud is capable of providing abundant resources for a large number of businesses. Therefore, the resource requirement of one single customer is generally easy to be met on Huawei Cloud. It can be assumed that resources are always available.

Quick resource acquisition

- Resources can be obtained in just a few minutes.
- Therefore, users can easily implement dynamic service capabilities. Obtaining resources for service innovation is also convenient.

No need for capacity guessing

- Resources can be quickly obtained, so they are there when needed.
- Due to the large-scale resources, it can be assumed that resources are always available.

Easier O&M

- More automated and less manual O&M
- More application-level and less infrastructure-level O&M

Operations at different geographical locations

- The same standardized services are used to provide infrastructure.
- API-based access allows resources to be obtained in the same way.

Agile services

- Compute resources are quickly obtained to support agile services.
- A solid foundation is laid for companies to go cloud native and promote business innovation.

Focus on the core value of businesses and avoid repeated work

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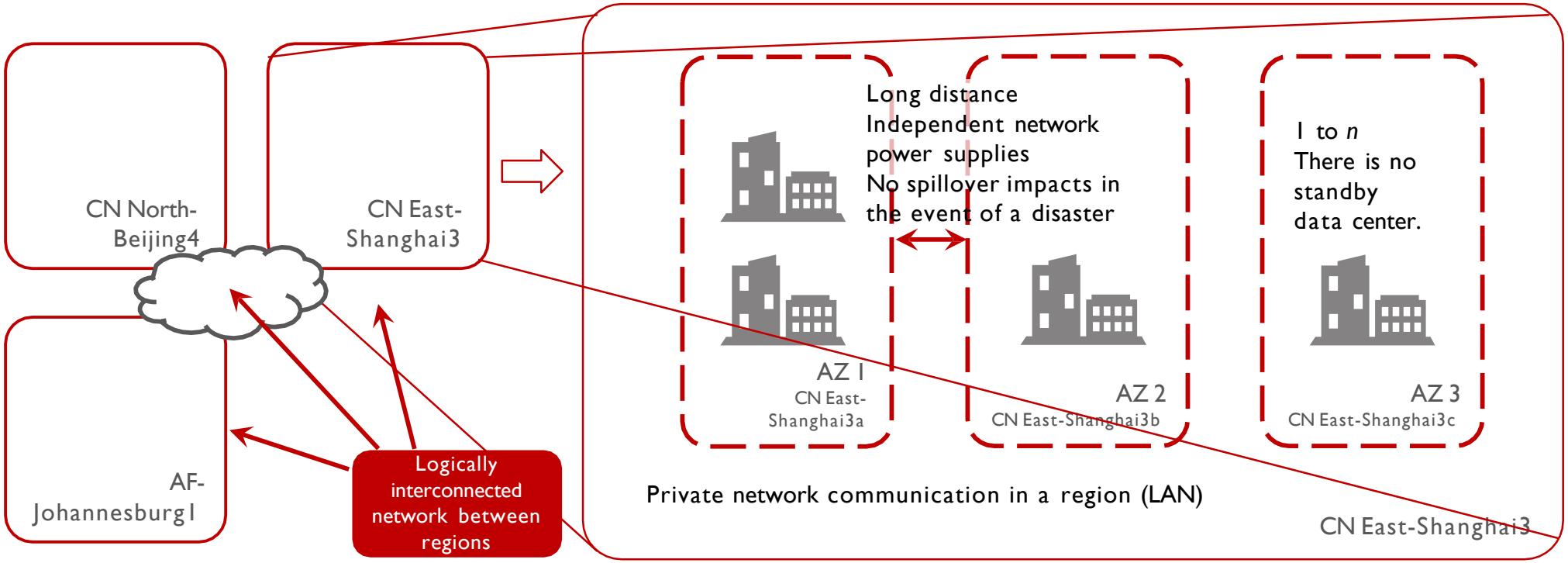
Huawei Cloud Infrastructure

- A region is a collection of resources divided by a geographic location. Each region comprises one or more AZs and is completely isolated from other regions. AZs within the same region can communicate with one another through a private network, while those in different regions cannot.
- An availability zone (AZ) is a geographical area with independent power supply and networks in the same service region. Generally, an AZ is an independent physical equipment room. One region has multiple AZs. If one AZ becomes faulty, the other AZs in the same region can still function. AZs within the same region can communicate with one another through a private network.



How Do I Select a Region?

01 Compliance is a must.

02 User experience comes first.

03 Functions are region-specific.

04 Cost savings are higher.

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How Do I Select an AZ?

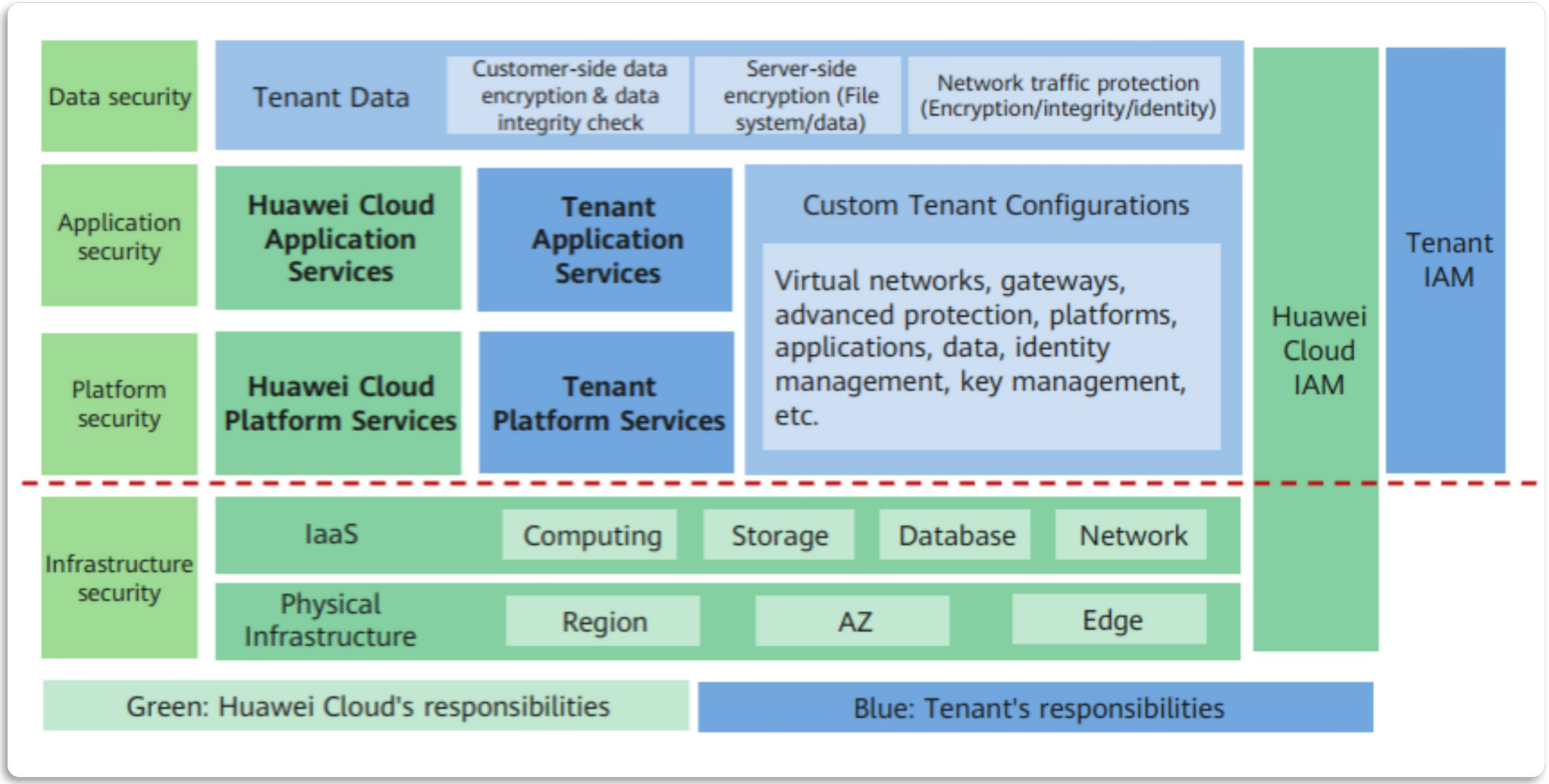
AZ

1. AZs aim to improve availability.
2. AZs help build active-active and multi-active applications.
3. AZs are equivalent.

Application requirements prevail in the selection.

If there is no special requirement, one AZ is sufficient.
If high availability (HA) is required, select at least two AZs.
One more AZ does not increase costs.
If there are more than three AZs, the availability improvement is limited and the complexity increases.

Huawei Cloud Shared Responsibility Model



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Huawei Cloud Security Certifications

Security compliance

- ISO 27001 (ISO/IEC 27001 Information Security Management)
- ISO 27017 (ISO/IEC 27017 Security Controls for Cloud Services)
- CSA STAR Certification (CSA STAR Cloud Security)
- ISO 20000 (ISO/IEC 20000-1 Information Technology Service Management)
- ISO 22301 (ISO 22301 Business Continuity Management)
- Singapore MTCS (SS 584 Multi-tiered Cloud Computing Security)
- SOC1 Type2
- SOC2 Type2
- SOC3

Privacy compliance

- ISO 27701 (ISO/IEC 27701 Privacy Information Management)
- ISO 27018 (ISO/IEC 27018 Code of practice for protection of personally identifiable information (PII) in public clouds acting as PII processors)
- ISO 29151 (ISO 29151 General personal information protection)
- BS 10012 (BS 10012 Data Protection)
- SOC2 Type1 Privacy
- Singapore PDPA
- PDPO of Hong Kong (China)
- Malaysia PDPA
- Thailand PDPA
- South Africa POPI

Financial compliance

- PCI DSS (PCI DSS Payment Card Industry Data Security Standard)
- PCI 3DS
- Singapore OSPAR
- PCI DSS Practice Guide
- Singapore MAS & ABS
- HKMA & SFC in Hong Kong (China)
- Malaysia BNM&SC
- Thailand BoT & OSEC

Healthcare compliance

- ISO 27799 (ISO 27799 Health Informatics)
- HIPAA

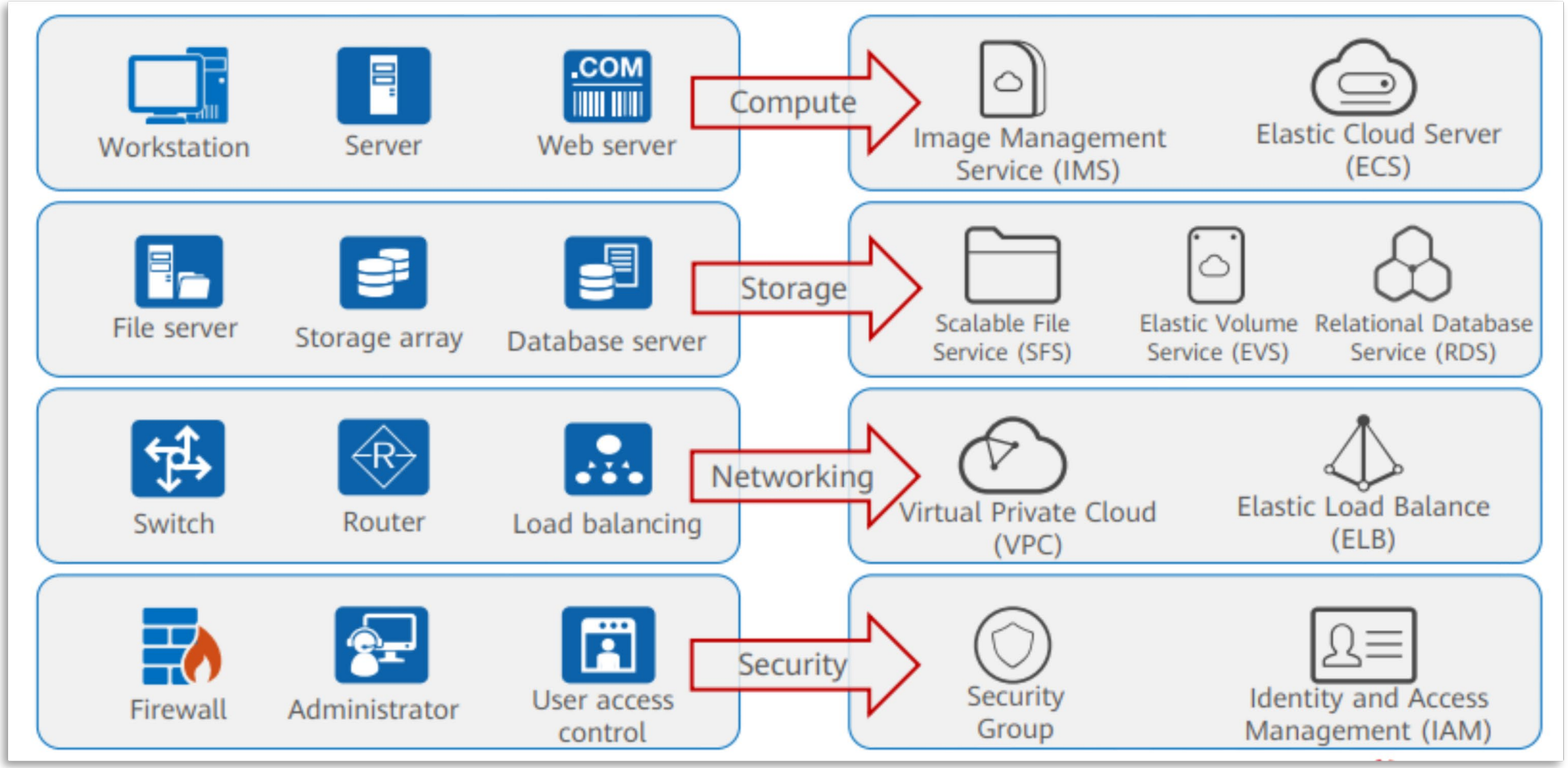
Compliance in China

- DJCP
  - Public cloud O&M system (level 3)
  - Public cloud operations system (level 3)
  - Public cloud service system (level 3)
  - Public cloud PaaS system (level 3)
  - Public cloud SaaS system (level 3)
- DJCP
  - Public cloud high-level protection service system (level 4)
  - Public cloud high-level protection PaaS system (level 4)
  - Public cloud high-level protection SaaS system (level 4)
- CAC (www.cac.gov.cn) — Cybersecurity Review by the Cyberspace Administration of China (CAC)
- ITSS — ITSS Cloud Computing Service Capability Assessment by the MIIT
- CUDPCA — Certification for the Capability of Protecting Cloud Service User Data
- TRUCS Gold O&M Assessment

Trusted cloud in China

- TRUCS 可信云服务评估 — Trusted cloud service assessment
  - Cloud server
  - RDS
  - Cloud cache
  - OBS
  - Block storage
  - Cloud distribution
  - Workspace
  - Cloud backup
  - Direct Connect
  - Content security
  - Message queue
  - Card OCR
  - GPU cloud servers
  - Physical cloud server
  - Cloud server security
  - Local load balancing
  - Security operation center
  - Situational awareness platform
  - Cloud-native databases
  - Distributed database middleware
- Trusted cloud solution
  - Container security
  - Content security
  - E-government cloud
  - Hybrid cloud
  - Trusted cloud container
  - Open-source solutions
  - Trusted e-government cloud
  - Micro-platform service
  - Cloud server classification
  - Hybrid cloud security
  - Cloud service provider credit rating
  - CDN credit rating
  - Shared responsibility model for security
  - Edge cloud trustworthiness
  - Cloud computing risk management capability
  - Cloud-edge collaboration management solution
- Big data product capability evaluation
  - Basic capabilities of the distributed batch processing platform
  - Performance evaluation of the distributed batch processing platform
  - Basic capability evaluation of distributed analytical databases
  - Time series databases
  - Database management tools
  - Knowledge graph
  - Basic capability evaluation of distributed transactional databases

Overview Cloud Services Components



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ECS Features

Elastic Cloud Server (ECS)

- Use an ECS just like a server.
- You have complete control over an ECS

Note: you only need to pay attention to the parts abrove to OS

- You can adjust computing resources after you buy an ECS
- Pay only for the resource you use

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Billing Mode

Pay-per-use

- On-demand use
- Pay-as-you-go
- Suitable for elastic or temporary use

Yearly/Monthly

- Like self-owned serves
- Monthly or yearly subscriptions
- Stable discount
- Suitable for long-term use

Spot Price

- The price is determined by the market demand
- Dynamic pricing but possibly high discounts
- Resources maty be reclaimed
- Suitable for architecture that allow dynamic resource adjustment

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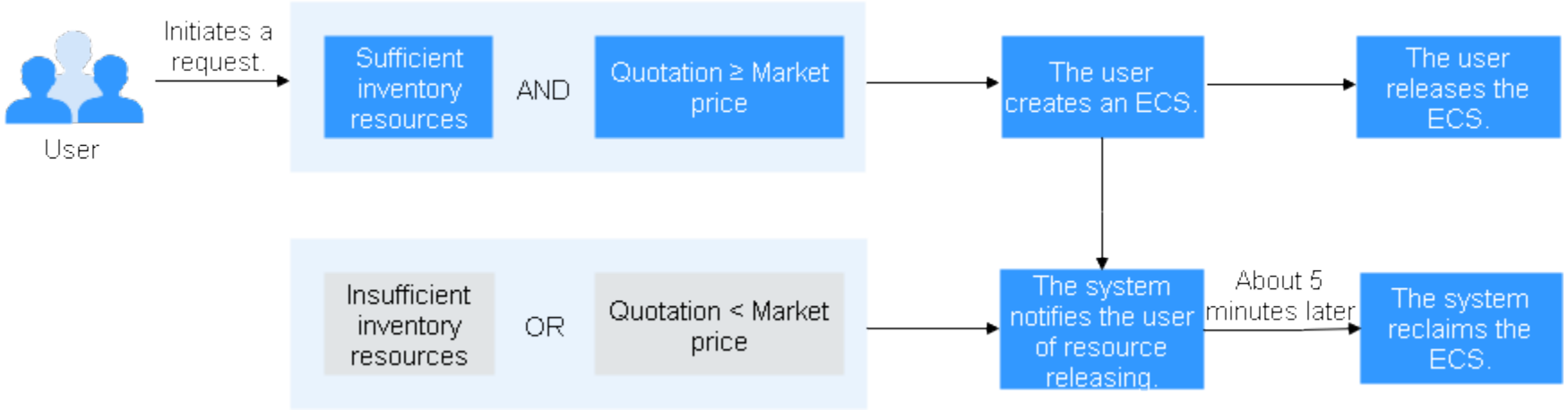
Purchasing Spot ECS

- If the maximum price is greater than or equal to the market price and the inventory resources are sufficient, **the spot ECS** can be purchased **and will be billed at the market price**.

- If the maximum price is less than the market price, the spot ECS cannot be purchased.



ECS Configuration

1 Basic Settings

- Billing Mode
- Region and AZ
- Instance type
- Boot Image
- EVS Disk

2 Network Settings

- VPC and Subnet
- Security group
- EIP and traffic

3 Advanced Setting

- ECS name
- Server backup
- Server group
- Advanced option
  Tag/user data/ etc.

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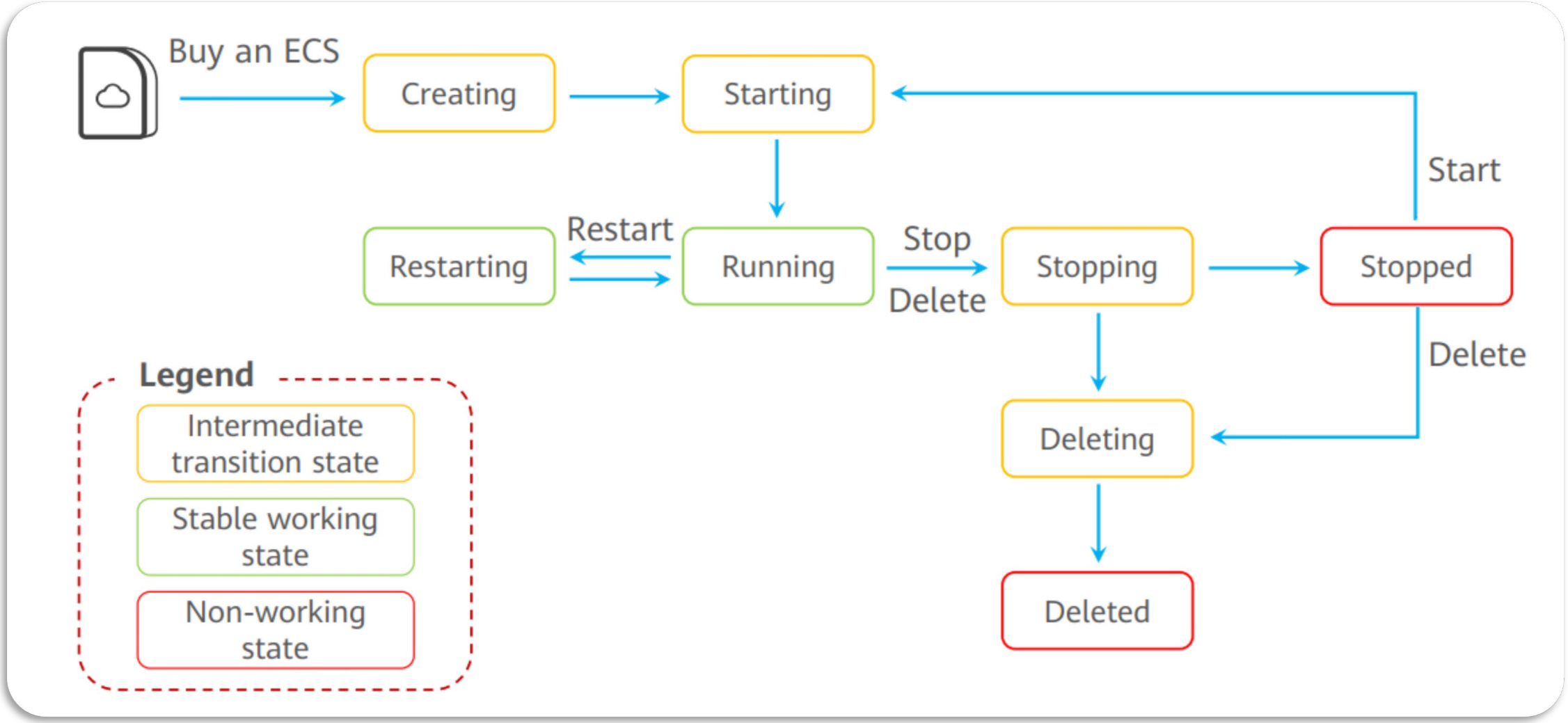
Sever life Cycle Management



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Specifications

C8.8xlarge.4

Instance family

General computing plus (c)
General computing (s)
memory-optimized (m)
etc.

Instance generation

Generally, the newer the generation, the better the cost-effectiveness.

Ratio of memory to vCPUs

Number of vCPUs

	vCPU	Memory
c7n.large.2	2	4
c7n.large.4	2	8
c7n.xlarge.2	4	8
c7n.xlarge.4	4	16
c7n.2xlarge.2	8	16
c7n.2xlarge.4	8	32
c8.8xlarge.4	32	128

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Dedicated Host & Bare Metal Server

Dedicated Host (DeH)

- Services are running on the ECSs deployed a DeH
- The physical resources of the DeH are not shared with others
- DeH meets the compliance requirements for exclusive use.
- DeH addresses hardware specification requirements of specified software license
- Customer can choose vCPU and Memory
- Customer can create ECS on DeH

Bear Metal Server (BMS)

- Scalability of VMs and high performance of physical servers
- Dedicated servers on the cloud
- Customer can choose physical core
- Cannot create ECS on BMS

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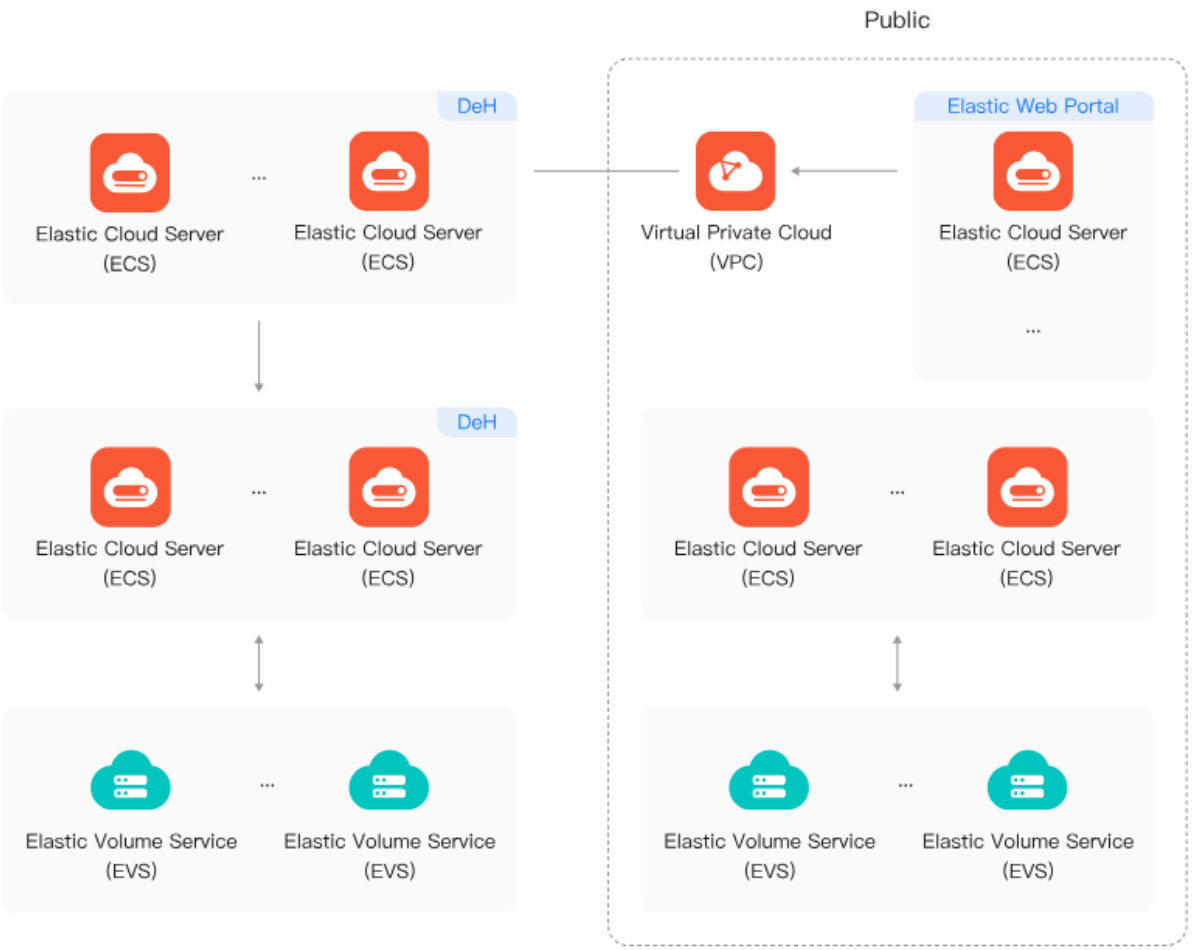
Dedicated Host (DeH)



Scenario

Systems and data need to be isolated from those in the public resource pool to ensure data security. During peak hours, more servers with stable performance are necessary.

Advantages

- Resource Isolation - Physical resources are isolated to ensure different user workloads can never impact each other and optimal performance is ensured.
- Connection to ECSs in Public Resource Pool - DeHs are deployed in the same VPC as the ECSs in the public resource pool and can directly communicate with these ECSs.

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IMS Features

Image Management Service (IMS)

- An image is a template of an ECS environment.
- An image specifies resource limits.
- An image contains an OS.
- An image can be used to restore an ECS.
- A variety of image types are provided by Huawei Cloud

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Image Type

Public Images

- All marketplace images are thoroughly tested before being published.
- Operating systems are **pre-installed in public images**. (Linux and Windows)
- Huawei Cloud periodically updates and maintains public images.
- A public image is a stable environment for customization.

Marketplace Images

- Marketplace images are **third-party images available in KooGallery**.
- All marketplace images are thoroughly test before being published.
- The billing is determined by image provider
- Specific applications are pre-installed in marketplace images.

Private Images

- You can **create a private image** from your ECS.
- You are responsible for the image security.
- When you create an image, your ECS should be stopped.
- You can **install custom software as needed**.

Shared Images

- You can **share your private images with other Huawei Cloud accounts**.
- You need to configure permissions before sharing.
- Image providers are responsible for the security of shared images.
- Ensure that a shared image is from a trusted sharer

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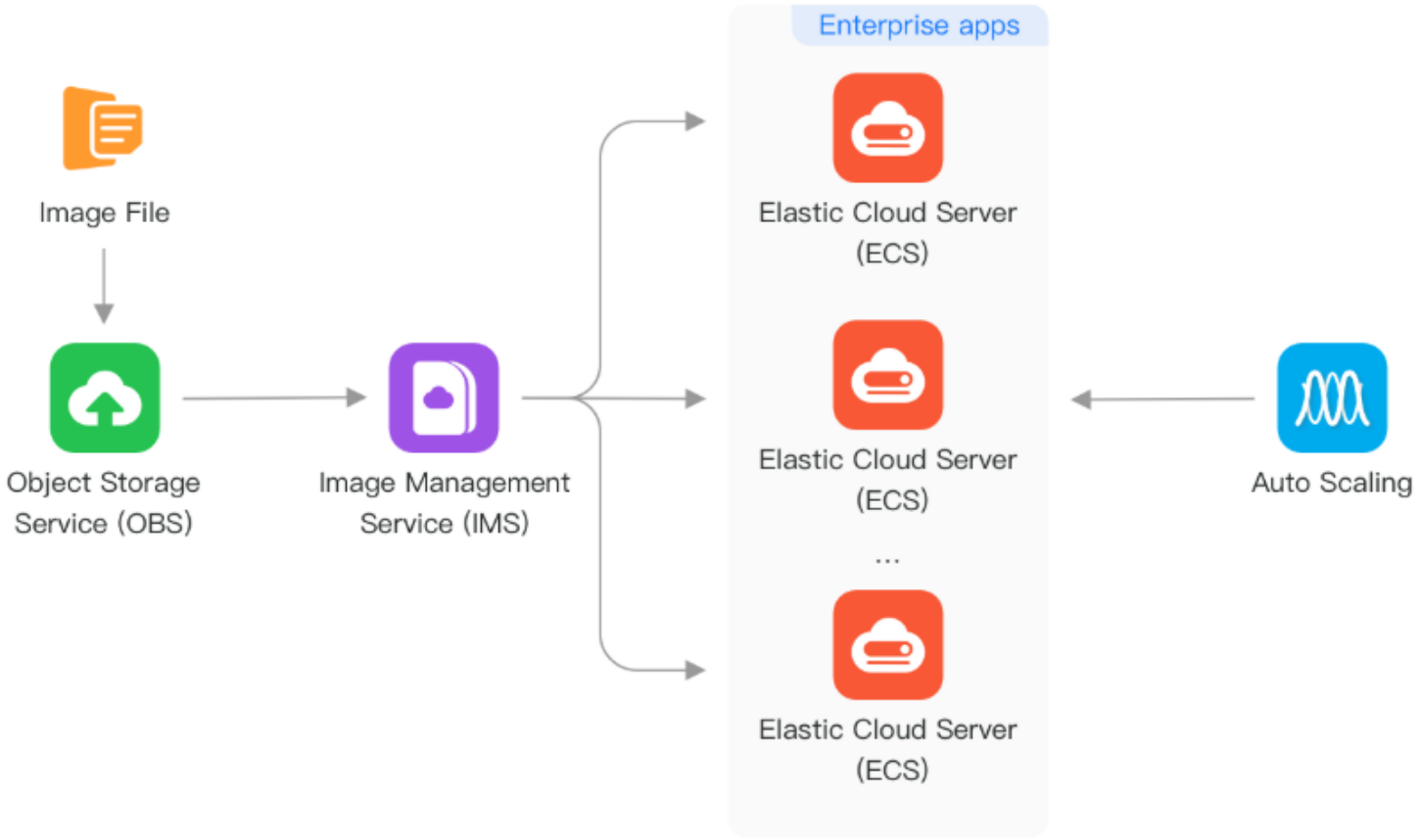
Image Management Service (IMS)



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Bring digital to every person, home and organization for a fully connected, intelligent world.

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